

/*****

Program Name: Table_14_2_1.sas

Project: Clinical SAS Project
Study : Adverse Events Analysis

Purpose : Generate Table 14.2.1 - Treatment Emergent Adverse Events
by Treatment, System Organ Class, and Preferred Term
for the Safety Population.

Input Data: ADSL
ADAE

Output: Table_14_2_1.rtf

Programmer: Nithish M S

Created Date: 20JUN2026

Software: SAS 9.4

Modification History

Date	Programmer	Description
20JUN2026	Nithish M S	Initial version

*****/

/* IMPORT ADSL */

```

PROC IMPORT DATAFILE="/home/u64252598/Project AE Table raw file Adverse_ADSL.xlsx"
  OUT=ADSL
  DBMS=XLSX
  REPLACE;
  GETNAMES=YES;
RUN;

```

Table: WORK.ADSL

View: Column names

Filter: (none)

Total rows: 8

Total columns: 26

Rows 1-8

	STUDYID	USUBJID	SUBJID	SITEID	COUNTRY	AGE	AGEU	AGEGR1	SEX	ETHNIC	RACE
1	009/01	A02	1	A	USA	45	YEARS	18 TO 60	M	HISPANIC OR LATINO	WHITE
2	009/01	A03	2	A	USA	52	YEARS	18 TO 60	M	NON HISPANIC OR LATINO	ASIAN
3	009/01	A01	3	A	USA	49	YEARS	18 TO 60	M	NON HISPANIC OR LATINO	WHITE
4	009/01	A01	4	B	USA	57	YEARS	18 TO 60	F	NON HISPANIC OR LATINO	WHITE
5	009/01	B04	5	B	USA	55	YEARS	18 TO 60	M	NON HISPANIC OR LATINO	ASIAN
6	009/01	B04	6	C	USA	59	YEARS	18 TO 60	F	HISPANIC OR LATINO	WHITE
7			.			.					
8			.			.					

/* SAFETY POPULATION */

DATA ADSL1;

SET ADSL;

IF SAFFL='Y';

IF INDEX(UPCASE(TRT01A),"100 MG TG3304")>0 THEN DO;

TRT='A';

ORD=1;

END;

ELSE IF INDEX(UPCASE(TRT01A),"PLACEBO")>0 THEN DO;

TRT='B';

ORD=2;

END;

KEEP USUBJID TRT ORD;

RUN;

/* BIG N */

PROC SQL NOPRINT;

SELECT COUNT(DISTINCT USUBJID)

INTO :N1-:N2

FROM ADSL1

GROUP BY ORD

ORDER BY ORD;

QUIT;

```
%PUT &=N1 &=N2;
~
71      SELECT COUNT(DISTINCT USUBJID)
72      INTO :N1-:N2
73      FROM ADSL1
74      GROUP BY ORD
75      ORDER BY ORD;
NOTE: The query as specified involves ordering by an item that doesn't appear in its SELECT clause.
76
77      QUIT;
NOTE: PROCEDURE SQL used (Total process time):
      real time      0.00 seconds
      user cpu time   0.01 seconds
      system cpu time 0.01 seconds
      memory          5606.18k
      OS Memory       27048.00k
      Timestamp       06/20/2026 03:18:48 PM
      Step Count      52      Switch Count  0
      Page Faults     0
      Page Reclaims   143
      Page Swaps      0
      Voluntary Context Switches 0
      Involuntary Context Switches 0
      Block Input Operations 0
      Block Output Operations 16

78
79      %PUT &=N1 &=N2;
N1=3 N2=2
80
81      OPTIONS NONOTES NOSTIMER NOSOURCE NOSYNTAXCHECK;
91
```

/* IMPORT ADAE */

```
PROC IMPORT DATAFILE="/home/u64252598/Project AE Table raw file Adverse_ADAE.xlsx"
OUT=ADAE
DBMS=XLSX
REPLACE;
GETNAMES=YES;
RUN;
```

e:

WORK.ADAE

 | View:

Column names

 | Filter: (none)

Total rows: 6 Total columns: 30

STUDYID	USUBJID	SUBJID	SITEID	AESQ	AETERM	AELLT	AELLTCD	AEDECOD	AEPTCD	A
009/01	A02	1	A	1	BLURRED VISION	BLURRED VISION	10037844	Blurry vision	10037844	R
009/01	A03	2	A	1	RASHES	RASHES	10037844	Skin Rashes	10037844	R
009/01	A01	3	A	1	NAUSEA	NAUSEA	10037844	Nausea	10037844	R
009/01	A01	4	B	1	NAUSEA	NAUSEA	10037844	Nausea	10037844	R
009/01	B04	5	B	2	HEADACHE	HEADACHE	10037844	Headache	10037844	R
009/01	B04	6	C	1	RASHES	RASHES	10037844	Skin Rashes	10037844	R

/* TEAE POPULATION */

```
DATA ADAE1;
SET ADAE;

IF SAFFL='Y' AND TEAEFL='Y';

IF INDEX(UPCASE(TRTA),"100 MG TG3304")>0 THEN DO;
  TRT='A';
  ORD=1;
END;
```

```
ELSE IF INDEX(UPCASE(TRTA),"PLACEBO")>0 THEN DO;
  TRT='B';
  ORD=2;
END;
```

RUN;

Table: WORK.ADAE1 | View: Column names | | Filter: (none)

Total rows: 6 Total columns: 32

Rows 1-6

1	TRT01SDT1	TRT02SDT1	TRT01SDTN	TRT02SDTN	ASTDT	AENDT	APERIOD	TRTA	TRT	ORD
19	10/22/2018	11/22/2018	21479	21510	12/13/2018	02/11/2019	2	PLACEBO	B	2
19	10/22/2018	11/22/2018	21479	21510	12/13/2018	02/15/2019	2	100 MG TG3304	A	1
19	10/22/2018	11/22/2018	21479	21510	11/03/2018	01/10/2019	1	100 MG TG3304	A	1
19	10/22/2018	11/22/2018	21479	21510	12/13/2018	02/11/2019	2	PLACEBO	B	2
19	10/22/2018	11/22/2018	21479	21510	11/03/2018	02/11/2019	1	PLACEBO	B	2
19	10/22/2018	11/22/2018	21479	21510	11/03/2018	02/16/2019	1	100 MG TG3304	A	1

/* OVERALL SUBJECTS WITH TEAES */

PROC SQL;

```
CREATE TABLE OVERALL AS
SELECT TRT,
  "Number of Subjects with TEAEs" AS AEBODSYS LENGTH=200,
  "" AS AEDECOD LENGTH=200,
  COUNT(DISTINCT USUBJID) AS N,
  0 AS LVL
FROM ADAE1
GROUP BY TRT;
```

Table: WORK.OVERALL | View: Column names | | Filter: (none)

Total rows: 2 Total columns: 5

	TRT	AEBODSYS	AEDECOD	N	LVL
1	A	Number of Subjects with TEAEs		3	0
2	B	Number of Subjects with TEAEs		3	0

/* SOC LEVEL */

```
CREATE TABLE SOC AS
SELECT TRT,
  AEBODSYS,
  "" AS AEDECOD LENGTH=200,
  COUNT(DISTINCT USUBJID) AS N,
  1 AS LVL
FROM ADAE1
```

GROUP BY TRT, AEBODSYS;

Table:

WORK.SOC

 | View:

Column names

 |     |  Filter: (none)

Total rows: 5 Total columns: 5

	TRT	AEBODSYS	AEDECOD	N	LVL	
1	A	Gastrointestinal signs and symptoms		1	1	
2	A	Skin subcutaneous Tissue Disorders		2	1	
3	B	Eye Disorders		1	1	
4	B	Gastrointestinal signs and symptoms		1	1	
5	B	Nervous system disorders		1	1	

/* PT LEVEL */

```
CREATE TABLE PT AS
SELECT TRT,
       AEBODSYS,
       AEDECOD,
       COUNT(DISTINCT USUBJID) AS N,
       2 AS LVL
FROM ADAE1
GROUP BY TRT, AEBODSYS, AEDECOD;
```

QUIT;

Table:

WORK.PT

 | View:

Column names

 |     |  Filter: (none)

Total rows: 5 Total columns: 5

	TRT	AEBODSYS	AEDECOD	N	LVL	
1	A	Gastrointestinal signs and symptoms	Nausea	1	2	
2	A	Skin subcutaneous Tissue Disorders	Skin Rashes	2	2	
3	B	Eye Disorders	Blurry vision	1	2	
4	B	Gastrointestinal signs and symptoms	Nausea	1	2	
5	B	Nervous system disorders	Headache	1	2	

/* STACK DATA */

```
DATA ALL;
SET OVERALL SOC PT;
RUN;
```

Table: WORK.ALL | View: Column names |     Filter: (none)

Total rows: 12 Total columns: 5

	TRT	AEBODSYS	AEDECOD	N	LVL
1	A	Number of Subjects with TEAEs		3	0
2	B	Number of Subjects with TEAEs		3	0
3	A	Gastrointestinal signs and symptoms		1	1
4	A	Skin subcutaneous Tissue Disorders		2	1
5	B	Eye Disorders		1	1
6	B	Gastrointestinal signs and symptoms		1	1
7	B	Nervous system disorders		1	1
8	A	Gastrointestinal signs and symptoms	Nausea	1	2
9	A	Skin subcutaneous Tissue Disorders	Skin Rashes	2	2
10	B	Eye Disorders	Blurry vision	1	2
11	B	Gastrointestinal signs and symptoms	Nausea	1	2
12	B	Nervous system disorders	Headache	1	2

/* SORT*/

```
PROC SORT DATA=ALL;
BY LVL AEBODSYS AEDECOD TRT;
RUN;
```

/* TRANSPOSE*/

```
PROC TRANSPOSE DATA=ALL OUT=TRANS;
BY LVL AEBODSYS AEDECOD;
ID TRT;
```

Table: WORK.TRANS | View: Column names |     Filter: (none)

Total rows: 9 Total columns: 6

	LVL	AEBODSYS	AEDECOD	_NAME_	A	B
1	0	Number of Subjects with TEAEs		N	3	3
2	1	Eye Disorders		N	.	1
3	1	Gastrointestinal signs and symptoms		N	1	1
4	1	Nervous system disorders		N	.	1
5	1	Skin subcutaneous Tissue Disorders		N	2	.
6	2	Eye Disorders	Blurry vision	N	.	1
7	2	Gastrointestinal signs and symptoms	Nausea	N	1	1
8	2	Nervous system disorders	Headache	N	.	1
9	2	Skin subcutaneous Tissue Disorders	Skin Rashes	N	2	.

VAR N;

RUN;

/* CREATE DISPLAY VALUES*/

DATA FINAL;

SET TRANS;

LENGTH TERM \$200

DRUGA \$30

DRUGB \$30;

/* Treatment A */

IF MISSING(A) THEN DRUGA='0';

ELSE IF A=&N1 THEN

DRUGA=CATS(PUT(A,3.),'(100%));

ELSE

DRUGA=CATS(PUT(A,3.),'(',
PUT((A/&N1)*100,5.1),'%));

/* Treatment B */

IF MISSING(B) THEN DRUGB='0';

ELSE IF B=&N2 THEN

DRUGB=CATS(PUT(B,3.),'(100%));

ELSE

DRUGB=CATS(PUT(B,3.),'(',
PUT((B/&N2)*100,5.1),'%));

/* Display hierarchy */

IF LVL=0 THEN

TERM='Number of Subjects with TEAEs';

ELSE IF LVL=1 THEN

TERM=AEBODSYS;

ELSE IF LVL=2 THEN

TERM=' ||STRIP(AEDECOD);

RUN;

/* REPORT */

ODS ESCAPECHAR='^';

ODS RTF FILE="/home/u64252598/Table_14_2_1.rtf"
STYLE=JOURNAL;

TITLE1 J=C H=6PT

"TABLE 14.2.1";

TITLE2 J=C H=6PT

"TREATMENT EMERGENT ADVERSE EVENTS BY TREATMENT, SYSTEM ORGAN CLASS
AND PREFERRED TERM";

TITLE3 J=C H=6PT

"(SAFETY POPULATION)";

PROC REPORT DATA=FINAL NOWD HEADLINE HEADSKIP SPLIT='|';

COLUMN TERM

("Treatment"

DRUGA

DRUGB);

DEFINE TERM / DISPLAY

"MedDRA@ System Organ Class|MedDRA@ Preferred Term"

STYLE(COLUMN)=[JUST=L CELLWIDTH=50%];

DEFINE DRUGA / DISPLAY

"100 MG TG3304|(N=&N1)"

STYLE(COLUMN)=[JUST=C CELLWIDTH=20%];

DEFINE DRUGB / DISPLAY

"Placebo|(N=&N2)"

STYLE(COLUMN)=[JUST=C CELLWIDTH=20%];

RUN;

ODS RTF CLOSE;

TITLE;

TABLE 14.2.1
TREATMENT EMERGENT ADVERSE EVENTS BY TREATMENT, SYSTEM ORGAN CLASS AND PREFERRED TERM
(SAFETY POPULATION)

MedDRA@ System Organ Class MedDRA@ Preferred Term	Treatment	
	100 MG TG3304 (N=3)	Placebo (N=2)
Number of Subjects with TEAEs	3(100%)	3(150.0%)
Eye Disorders	0	1(50.0%)
Gastrointestinal signs and symptoms	1(33.3%)	1(50.0%)
Nervous system disorders	0	1(50.0%)
Skin subcutaneous Tissue Disorders	2(66.7%)	0
Blurry vision	0	1(50.0%)
Nausea	1(33.3%)	1(50.0%)
Headache	0	1(50.0%)
Skin Rashes	2(66.7%)	0